Local learning in networks of patch-clamp-fitted neurons needs each cell's adaptation in its eligibility trace

Author list to be confirmed by the authors

Preprint draft, November 9, 2026

Abstract

Synapses must assign credit across time using only local information, yet local learning rules are usually studied in networks of generic model neurons. We built recurrent networks entirely from differentiable firing-rate models fitted to patch-clamp recordings of mouse cortical cells. We kept each cell's intrinsic dynamics fixed and trained only the synapses, on three working-memory tasks. A cell-local, forward-in-time rule (e-prop with random feedback) was compared with exact back-propagation through time and with a frozen recurrent reservoir. All tests were preregistered and run on a held-out cohort of cells from different donor animals (16 paired networks per condition, 1,680 networks).

Nearly all fitted cells show spike-frequency adaptation. Removing the cell's own spike-history term from the eligibility trace lowered accuracy at unseen delays by 7–17 percentage points (significant in 5 of 6 task/optimizer settings), to the level of a frozen reservoir. Keeping the term preserved the alignment of the local update with the exact gradient (cosine up to 0.94), and the gain grew with each cell's adaptation strength. The local rule still trailed exact gradients by 11–22 points on tasks that combine memory with a later cue. This gap reflects ignored network paths rather than random feedback.

Once operating points were matched, cell identity and type-level wiring had no detectable effect; without matching, shuffled networks fired twice as fast. Cellular heterogeneity helped exact gradient learning by up to 25 points but did not help the local rule. Biologically plausible credit assignment in realistic circuits thus depends on synaptic traces that follow each neuron's measured adaptation.

1 Introduction

Learning a task that spans hundreds of milliseconds requires assigning credit to synapses whose influence on behaviour is delayed and passes through recurrent dynamics. The exact solution, back-propagation through time (BPTT; Werbos, 1990), needs information that a synapse does not have: the future error and its path through every other neuron. Local alternatives keep, at each synapse, an *eligibility trace* of how the synapse could have changed its postsynaptic cell. They combine it with a broadcast error or reward signal, in line with three-factor plasticity (Gerstner et al., 2018). Examples are RFLO (Murray, 2019) and e-prop (Bellec et al., 2020) (see also Williams and Zipser, 1989; Marschall et al., 2020). These rules are exact for a cell's own dynamics, and they approximate the rest of the network, for example with random feedback (Lillicrap et al., 2016).

What the trace must contain depends on the cell's own dynamics, and real cortical neurons are not the leaky units used in most such work. Their responses depend on several timescales, and most show spike-frequency adaptation (Benda and Herz, 2003). For spiking neurons with a single adaptive threshold, adaptation is known to support memory over behavioural timescales (Bellec et al., 2018; Salaj et al., 2021), and e-prop includes the adaptation variable in its eligibility trace (Bellec et al., 2020). Whether this matters for neurons whose intrinsic dynamics were *fitted to recordings*, with many heterogeneous filters per cell, has been hard to test: data-fitted single-cell models were usually not differentiable (Teeter et al., 2018), and large data-constrained network models have been trained with BPTT (Billeh et al., 2020; Chen et al., 2022). A recent bank of

differentiable generalized firing-rate (GFR) models, fitted to more than a thousand patch-clamp recordings (Yeung et al., 2025), now makes the test possible.

We built recurrent networks entirely from these fitted cells, keeping each cell's intrinsic parameters frozen and making only the synaptic weights plastic. We asked three questions: (i) how close a cell-local, forward-in-time rule comes to the exact gradient on working-memory tasks, compared with a frozen recurrent reservoir; (ii) whether the rule must include each cell's fitted spike-history filter in its eligibility trace; and (iii) whether the specific identity of the cells (the natural pairing of each cell's parameters and their heterogeneity) or the type-based wiring matters for learning, once confounds are removed.

The third question carries a trap that we found in our own earlier analyses. Shuffling parameters across cells moves each cell away from the operating point it was calibrated to. In our networks this roughly doubled firing rates and multiplied saturation, so an apparent effect of "destroying biological identity" could not be separated from a change of operating point. We therefore recalibrate every manipulated bank with the same label-free procedure before comparing. We also split cells by donor animal, compare learners at learning rates tuned on a separate development cohort, treat the network as the unit of inference (Aarts et al., 2014), fit every population-analysis axis on trials independent of those it is evaluated on (Kriegeskorte et al., 2009), and test all hypotheses in a protocol locked before the held-out confirmation cohort was touched.

Four things came out of the held-out confirmation (1,680 networks).

- The local rule needs each cell's fitted spike-history term. Without it, accuracy at unseen delays dropped by 7–17 pp, to about the level of a frozen reservoir. With it, the local estimate stayed aligned with the exact gradient, and the gain grew with each cell's adaptation strength.
- The local rule still trails BPTT on tasks that combine memory with a later cue. The gap comes from ignoring paths through other cells, not from random feedback.
- Cell identity and type-level wiring had no detectable effect once operating points were matched. Heterogeneity helped BPTT strongly but not the local rule.
- The operating point shaped local learning in a task-dependent way.

2 Results

2.1 Networks of fitted cells, three working-memory tasks, a locked protocol

Every network contains 100 GFR cells drawn from the confirmation cohort (80 Pyr, 10 PV, 5 SST, 5 VIP; Fig. 1). Each cell keeps its fitted input filters, spike-history filters and nonlinearity, with time constants from $\sim$0 to 2 s. Nearly every cell's history filter is net negative, i.e. adapting. Adaptation is strongest in pyramidal cells and weakest in PV cells (Fig. 1d,e). Networks learned three tasks. In the context-dependent decision (T1-early) the cue arrives with the stimulus. In T1-late it arrives only after the delay. In delayed match-to-sample (T2-dms) no input before the post-delay probe carries information about the choice. After 3000 updates every learner was near ceiling on in-distribution trials (Supplementary Fig. 8), so all comparisons below use the preregistered primary endpoint: accuracy on conflict trials at unseen, longer delays (400–600 ms). All results come from 1,680 runs on the held-out donor cohort (16 paired networks per condition), analysed exactly as locked before those runs (Supplementary Section A.1).

2.2 The local rule needs the cells' own spike-history term

Removing only the spike-history term β from the eligibility trace (Eq. 5) lowered unseen-delay conflict accuracy in all six task-by-optimizer comparisons, by 7–17 pp (Table 1, Fig. 2). The difference was significant after Holm correction in five of them. With Adam, accuracy fell from 81.6% to 70.6% (T1-early), from 75.7% to 66.5% (T1-late) and from 80.5% to 68.1% (T2-dms).

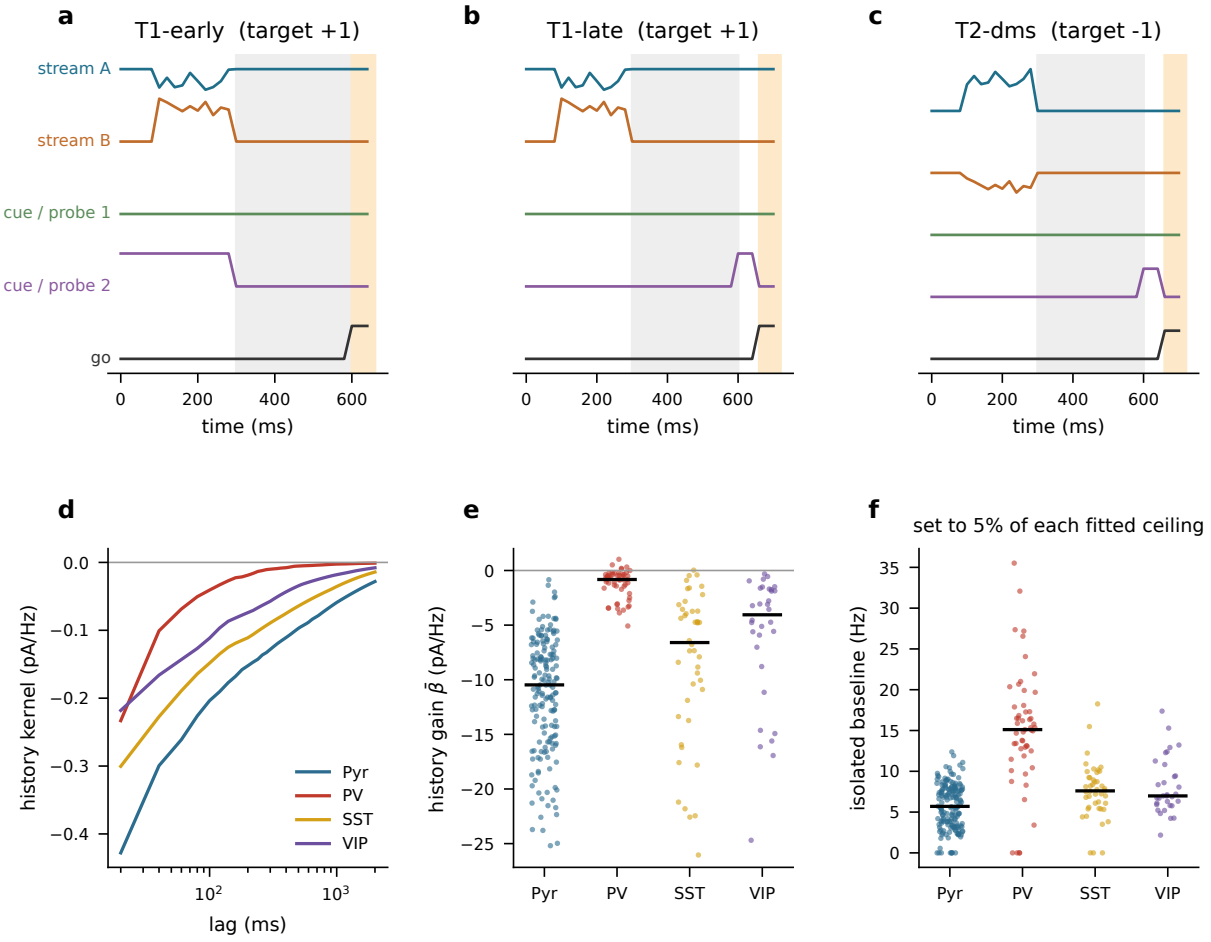


Figure 1: Tasks and data-fitted cells. (a–c) One conflict trial per task. Shading marks the delay (grey) and the response window (orange). (d) Median spike-history kernel by type, $\frac{1}{K}\sum_k \beta_{ik}(1-d_k)^{\ell-1}$. (e) Steady-state history gain $\bar{\beta}_i$ per cell; negative values mean adaptation. (f) Isolated baseline rate after calibration, set to 5% of each cell's fitted ceiling; the type hierarchy follows from the ceilings. Development cohort.

Without β, the rule's accuracy was close to or below that of a frozen reservoir that learns only its readout (descriptive). The one comparison that did not survive correction was T2-dms with RMS normalization (+7.1 pp, 95% CI [1.9, 11.9], Holm $p = 0.10$).

The local rule with β beat the frozen reservoir in all six comparisons. The gain was significant in three: T1-late with RMS and T2-dms with either optimizer (+10.6 to +11.9 pp). It was not significant on T1-early, where the reservoir already reached 76–77%, nor on T1-late with Adam (+6.1 pp). BPTT stayed clearly better than the local rule on the two tasks that require combining stored information with a later cue: +16.2 and +11.5 pp on T1-late, +14.9 and +22.3 pp on T2-dms (Adam, RMS). On T1-early the two did not differ.

2.3 Why: β aligns the local estimate with the true gradient

The local estimate differs from the exact gradient in two ways. It ignores paths through other cells, and it uses a random rather than the true feedback vector. Running the same rule with symmetric feedback isolates the first error. In networks trained with the local rule, the symmetric-feedback estimate with β kept a cosine of 0.76–0.94 with the exact recurrent gradient at the end of training. Without β the cosine was 0.54–0.66 (Fig. 3a–c). Random feedback lowered both curves, and β still helped (Fig. 3e). Per postsynaptic pyramidal cell, the cosine with symmetric feedback was 0.74 with β and 0.54 without. The improvement from β grew with the cell's adaptation loop gain

Table 1: Family A (confirmation cohort, 16 paired networks). Difference in unseen-delay conflict accuracy (pp) with 95% bootstrap CI; stars mark Holm-adjusted significance over all 18 tests ($^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$).

task	optimizer	A1: local − no β	A2: local − reservoir	A3: BPTT − local
T1-early	Adam	+10.9 [5.2, 16.8]*	+4.7 [-1.0, 11.0]	+4.0 [-1.7, 9.0]
T1-early	RMS	+16.8 [11.1, 22.6]**	+7.9 [2.6, 13.4]	-5.3 [-10.3, -0.4]
T1-late	Adam	+9.2 [5.2, 13.2]*	+6.1 [0.4, 12.0]	+16.2 [9.7, 21.0]**
T1-late	RMS	+10.6 [7.2, 14.1]**	+11.9 [7.2, 17.1]**	+11.5 [4.6, 16.7]*
T2-dms	Adam	+12.5 [9.0, 16.0]***	+10.6 [5.6, 15.3]*	+14.9 [11.4, 18.0]***
T2-dms	RMS	+7.1 [1.9, 11.9]	+10.7 [7.0, 14.5]***	+22.3 [17.5, 27.2]***

$|\langle g' \rangle \bar{\beta}|$. The per-network correlation was $r = 0.16$ [0.09, 0.23] (T1-early), 0.23 [0.16, 0.30] (T1-late) and 0.18 [0.11, 0.24] (T2-dms), across 16 networks each (Fig. 3d).

Behaviour followed the same split. With symmetric feedback, removing β still cost 8.4–12.7 pp (Family C). Replacing random by symmetric feedback gained only 3.8–4.4 pp (not significant). With symmetric feedback and β, the local rule matched BPTT on T1-early (−0.4 pp). On T1-late and T2-dms it still trailed by 12.4 and 11.0 pp. In this setting, therefore, most of the cost of ignoring a cell's adaptation lies in the eligibility trace itself, while the residual gap to BPTT on the harder tasks comes from cutting paths through other cells, not from random feedback.

2.4 Memory sits in the cells; learning shapes the response

Held-out decoders read the sign of *both* input streams from the delay-period population at ≥98% accuracy, for every learner including the frozen reservoir (Fig. 4). This held in T2-dms, where the second stream is a distractor. As designed, the choice could not be decoded in the T2-dms delay (49–50%). Learners differed in the response phase, where stored information must be combined with the cue. In T2-dms, response-phase choice decoding was 99.8% (BPTT), 98.9% (local), 92.5% (no β) and 83.3% (reservoir). The slow fitted filters are necessary for the memory: setting all time constants to ≤20 ms (with the static response preserved) dropped every task to chance (49–50%; −26 to −32 pp; Family C).

2.5 Cell identity and wiring do not matter once the operating point is matched

In the preregistered factorial, keeping each cell's own parameter combination rather than shuffling it within type changed the primary endpoint by +1.1 to +4.5 pp (Table 2, Fig. 5). No effect was significant. Four of six effects were practically negligible (90% CI within ±5 pp), and two were inconclusive (T1-late and T2-dms with RMS). The type-prior wiring versus degree-preserving rewiring was practically negligible in all six comparisons. No identity × learner or wiring × learner interaction was significant.

The operating point is what made earlier versions of this comparison misleading. With the intact cells' calibration kept ("reference"), shuffled networks fired at 24 Hz instead of 11–12 Hz, and 7% instead of 0.3–0.5% of their samples were near a cell's ceiling (Fig. 5d–f). At the tuned budget this did not change accuracy significantly (+1.8 to +3.3 pp versus matched shuffles). It does, however, change the regime being compared. Permuting whole parameter sets across types, with recalibration, cost 1.6–8.1 pp; only T2-dms reached nominal significance ($p = 0.003$, descriptive).

Heterogeneity helps the exact gradient but not the local rule. Replacing every cell by its type mean (recalibrated) lowered BPTT accuracy on T1-late by 12.6 pp with Adam (Holm $p \leq 0.001$) and on T2-dms by 20.1 and 24.9 pp. Under the local rule, the same manipulation had no significant effect in any comparison (−6.0 to +1.7 pp). On T1-early neither learner gained from heterogeneity. Type-mean networks also fired more slowly (7–8 Hz).

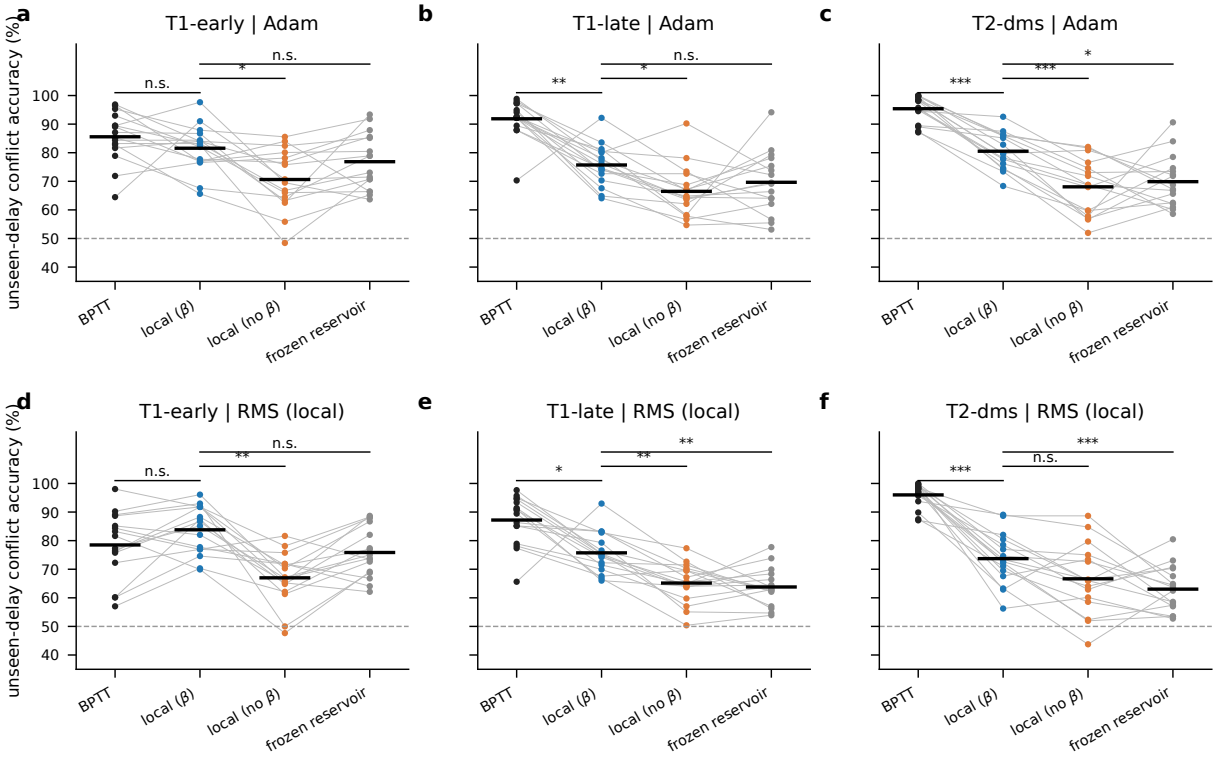


Figure 2: Learning rules (Family A). Unseen-delay conflict accuracy for each of 16 paired networks (dots; grey lines join the same network), with the mean (bar). Brackets: Holm-adjusted tests A3 (BPTT vs local), A1 (local with vs without β) and A2 (local vs frozen reservoir); n.s. = not significant.

2.6 The operating point shapes local learning, task by task

Moving every cell's calibrated baseline from 2% to 20% of its ceiling raised mean rates from about 7 Hz to 31–35 Hz (Fig. 6). BPTT was insensitive: no slope differed from zero. The local rule was not. Its accuracy fell with higher baselines on T1-early (-4.6 pp per doubling, 95% CI [-5.9, -3.3], $p < 0.001$), rose on T1-late ($+3.5$ pp, [1.6, 5.6], $p = 0.004$) and tended to rise on T2-dms ($+1.8$ pp, [0.0, 3.5], $p = 0.08$). No single operating point was best for local learning across tasks.

3 Discussion

We trained recurrent networks made entirely of data-fitted neuron models with a cell-local, forward-in-time learning rule, and tested every comparison in a protocol locked before the held-out cohort was used. Four results stand out.

The eligibility trace must contain the fitted adaptation. Almost every fitted cell adapts. A local rule whose eligibility trace ignores the cell's own spike-history filter lost 7–17 pp of out-of-distribution accuracy and fell to the level of a frozen reservoir. The reason is credit assignment: with β, the local estimate stays closely aligned with the exact gradient (cosine up to 0.94 with symmetric feedback), and the benefit is larger at cells with stronger adaptation loop gain. This extends a point made for single-threshold adaptive spiking neurons (Bellec et al., 2020, 2018; Salaj et al., 2021): it holds for heterogeneous, multi-timescale history filters fitted to real cells. For a synapse, the requirement is concrete. Its trace must decay with the postsynaptic cell's own adaptation kinetics, not only with membrane or synaptic time constants.

Where local learning falls short. On the two tasks that require combining stored information with a later cue, the local rule trailed BPTT by 11–22 pp. Symmetric feedback did not close that

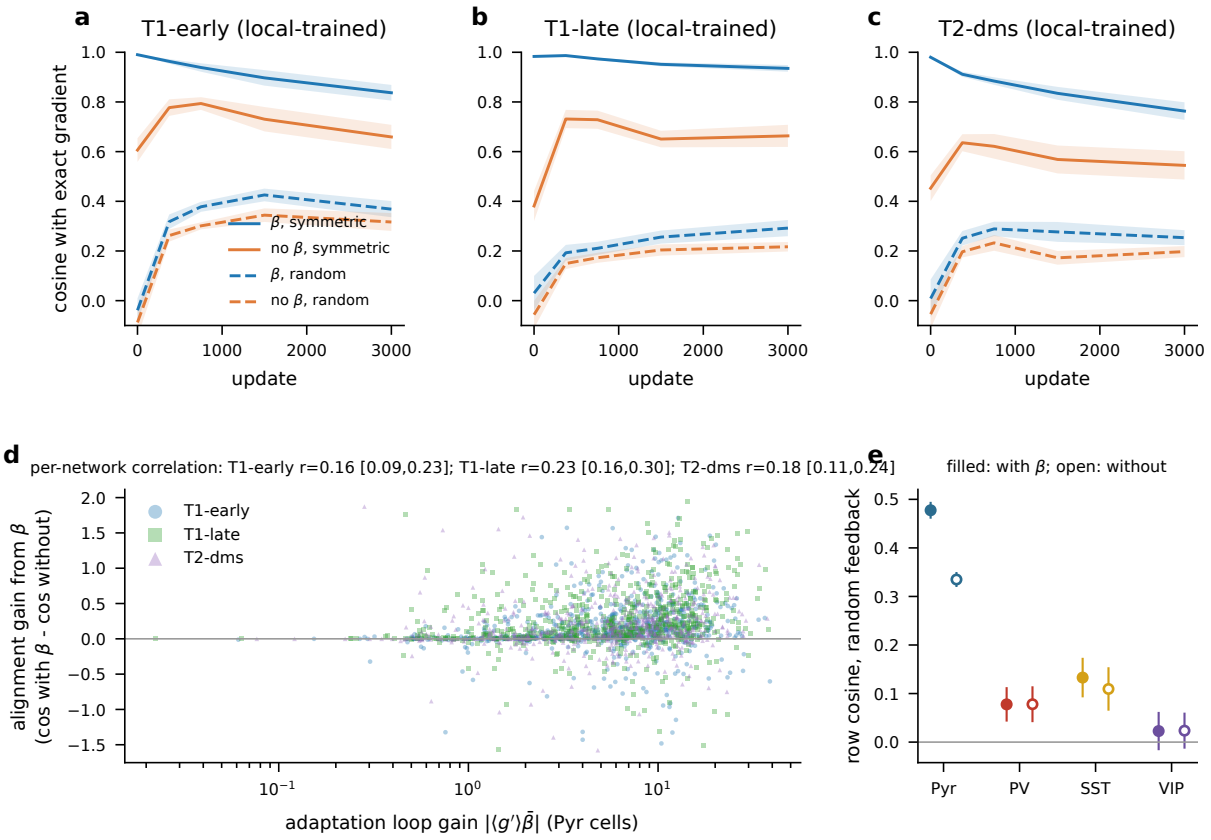


Figure 3: Mechanism. (a–c) Cosine between local estimates and the exact recurrent gradient over training, in networks trained with the local rule (Adam; mean ± s.e.m. over 16 networks). Solid: symmetric feedback; dashed: random feedback. Blue: with β; orange: without. (d) Per pyramidal cell: alignment gain from β (symmetric feedback) against the cell's adaptation loop gain, at the selected checkpoint. Title: per-network correlations, mean and 95% CI. (e) Mean row cosine by postsynaptic type with random feedback. Interneuron inputs receive no direct readout signal, so they align poorly either way.

gap, which points to the paths through other cells that the rule ignores, rather than to random feedback. The frozen reservoir was a strong baseline. Population codes show why: the slow fitted filters keep both input streams decodable through the delay in every network, so learning matters mainly in the response phase. A frozen reservoir is therefore the right control for claims about recurrent plasticity in networks of slow cells; in our data, local plasticity beat it significantly in half of the comparisons.

Biological specificity: a negative result with one exception. Once each manipulated bank was recalibrated to the same operating point, the natural pairing of each cell's parameters and the type-level wiring had no detectable effect. Most such effects were bounded within ±5 pp. Without recalibration, shuffled networks ran at twice the rate with about 15 times more saturation. That is a different regime, and earlier versions of this project had read differences between such networks as effects of cell identity. Heterogeneity was the exception. Replacing cells by their type means cost BPTT up to 25 pp on the harder tasks, in line with reports that heterogeneity aids learning (Perez-Nieves et al., 2021), but it did not affect the local rule. One possible reading is that the diversity of timescales is only useful to a learner that can assign credit through the network. A local rule that cannot do so does not benefit from it. Type-mean networks also fired more slowly, so part of this effect may be an operating-point difference that our calibration did not remove.

Operating point. The calibrated baseline rate, a choice that every data-constrained rate model makes, mattered for the local rule in a task-dependent way and not for BPTT. It should be reported

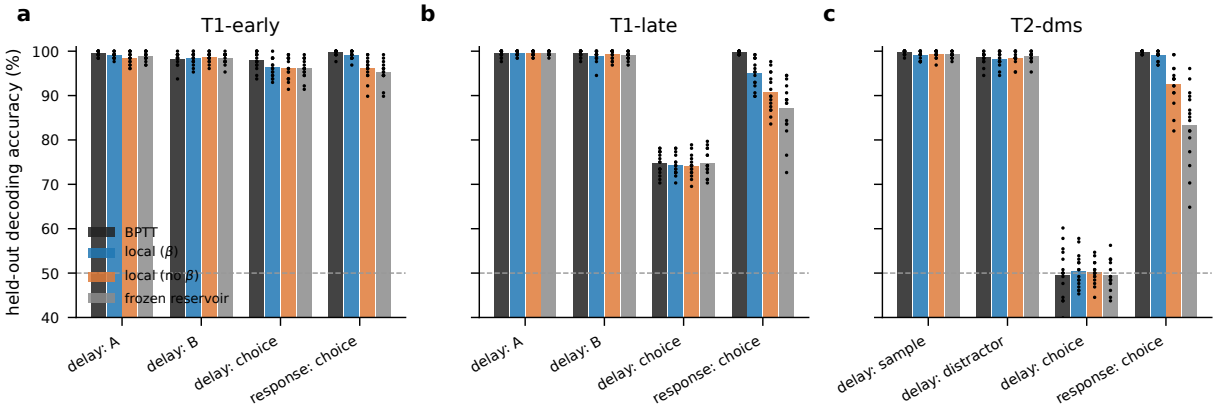


Figure 4: Population codes (Adam). Held-out decoding accuracy of each stream's sign and of the choice in the delay, and of the choice in the response phase. Decoders were fit on independent trials. Dots: networks.

Table 2: Family B. Identity: intact − within-type shuffle (matched); wiring: type prior − degree-preserving rewiring (factorial main effects, averaged over learner and the other factor); heterogeneity: intact − type-mean cells, for each learner. pp with 95% CI; stars: Holm within each task/optimizer family of 9. N: practically negligible (90% CI within ±5 pp); I: inconclusive; D: different.

task	opt.	identity	wiring	heterog., BPTT	heterog., local
T1-early	Adam	+2.3 [-1.0, 5.5] N	+1.1 [-0.6, 2.9] N	-0.3 [-4.8, 4.1] N	-3.0 [-7.5, 1.7] I
T1-early	RMS	+1.1 [-2.3, 4.3] N	-0.4 [-2.6, 1.7] N	-7.2 [-13.9, -0.9] I	-0.3 [-4.0, 3.3] N
T1-late	Adam	+1.9 [-1.1, 4.9] N	-0.2 [-2.8, 2.5] N	+12.6 [8.5, 16.4]*** D	-6.0 [-10.0, -1.0] I
T1-late	RMS	+4.5 [1.9, 7.2] I	-0.4 [-2.9, 1.9] N	+7.9 [2.8, 12.7] I	-2.3 [-6.6, 2.2] I
T2-dms	Adam	+1.3 [-2.3, 4.7] N	+0.2 [-1.8, 2.5] N	+20.1 [15.3, 25.0]*** D	+1.7 [-2.3, 5.6] I
T2-dms	RMS	+3.3 [-0.6, 7.4] I	+1.3 [-1.1, 3.9] N	+24.9 [17.7, 32.5]*** D	-5.7 [-11.4, -0.1] I

and varied, not fixed silently.

Limitations.

- *The models are rate models.* The GFR models reproduce each cell's firing-rate response, not spikes, dendrites or conductance-based synapses. The adaptation we find to matter is the fitted rate-history filter. Spike-level adaptation mechanisms are not modelled.
- *The cells and the tasks come from different areas.* The cells are mostly from mouse visual cortex, and the tasks are abstract working-memory tasks. We make no claim that these circuits solve these tasks in vivo.
- *The wiring is a prior, not data.* Type-level connection probabilities are modelling assumptions (Table 4). Measured probabilities were not used because the numbers we could trace were available only as figures.
- *The fits have a known split caveat.* Upstream, one stimulus protocol contributes to both fitting and validation. We selected cells on the released validation criterion and claim no new patch-clamp validation.
- *Optimization is a confound we only bound.* Adam with global clipping is not local. We therefore repeated every learning-rule comparison with per-synapse RMS normalization, which is local, but no optimizer is a model of biological plasticity. The local rule also uses a broadcast error and fixed random feedback; we did not test reward-based or neuromodulatory variants.

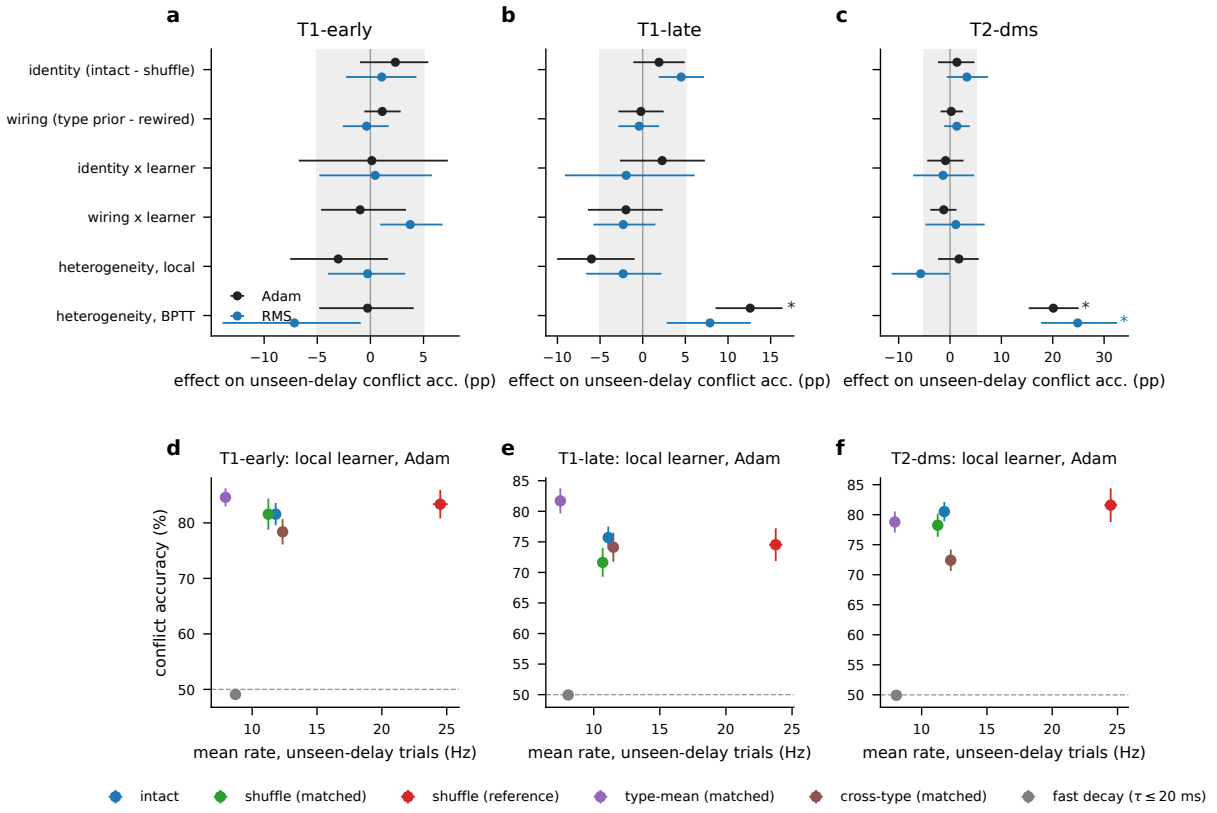


Figure 5: Biological specificity (Families B, C). (a–c) Preregistered effects with 95% CIs (black: Adam; blue: RMS). Grey band: ±5 pp; stars: Holm $p < .05$. (d–f) Mean rate and accuracy for the local learner under each manipulation (mean ± s.e.m.). The reference-calibrated shuffle doubles the rate and raises saturation about 15-fold, without a significant change in accuracy.

- *Scale.* Networks have 100 cells, with 16 paired seeds per condition. Each network uses 80 of 170 pyramidal cells, so networks share cells: inference is about networks built from this cohort, not about new cells.
- *The operating point is a modelling choice.* Calibrating every cell to a fixed fraction of its own ceiling sets the type-specific rate hierarchy by construction. Family D shows how much results depend on this choice.

Outlook. The same pipeline can test spiking versions of the fitted cells, measured connectivity once it is available in tabular form, reward-modulated rather than supervised signals, and larger networks. The protocol, code and all run outputs are available for reuse.

4 Methods

4.1 Data-fitted neuron models

We used the differentiable generalized firing-rate (GFR) models of Yeung et al. (2025). They were fitted to whole-cell current-clamp recordings in the Allen Cell Types Database (Gouwens et al., 2019), and we took the code and data from upstream commit `835a535`. Each cell i has $K = 8$ filters with fixed time constants $\tau_k \in \{\approx 0, 20, 50, 100, 200, 500, 1000, 2000\}$ ms. At the fitted bin size $\Delta t = 20$ ms the decay coefficients are $d_k = 1 - e^{-\Delta t/\tau_k}$, shared by all cells. With input current

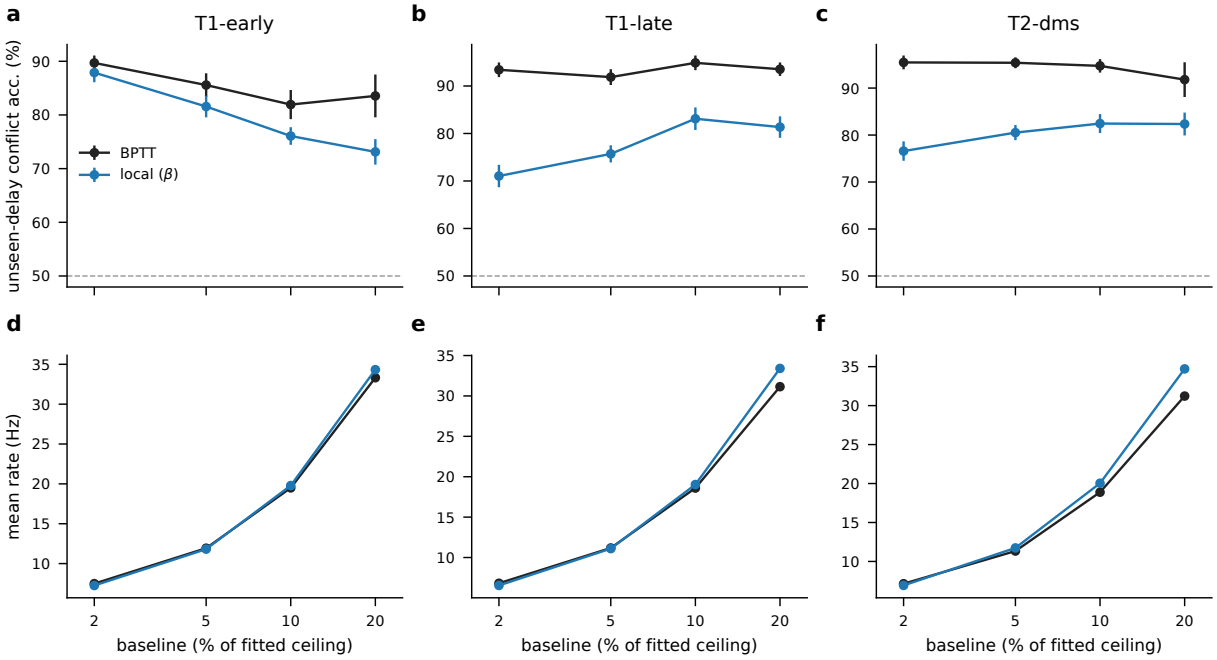


Figure 6: Operating point (Family D, descriptive). Unseen-delay conflict accuracy (a–c) and mean rate (d–f) as the calibrated baseline varies from 2% to 20% of each cell's fitted ceiling. Adam; mean ± s.e.m. over 16 networks.

$I_{i,t}$ (pA) and rate $r_{i,t}$ (Hz),

$$h_{i,t}^{(k)} = (1 - d_k)\, h_{i,t-1}^{(k)} + a_{ik}\, I_{i,t} + \beta_{ik}\, r_{i,t-1}, \tag{1}$$

$$r_{i,t} = \gamma_i\, \mathrm{ReLU}\Big[\tanh\Big(\textstyle\sum_d c_{id}^2\, x_{i,t}^d\Big)\Big], \qquad x_{i,t} = \big(\bar{h}_{i,t} - b_i\big)/\sigma_i, \quad \bar{h}_{i,t} = \tfrac{1}{K}\textstyle\sum_k h_{i,t}^{(k)}. \tag{2}$$

The stored a_{ik} and β_{ik} already include the discretization, so no extra Δt factor is applied. Upstream rates in spikes/ms are converted to Hz once. Fitted parameters are never changed by training. Our implementation reproduces the upstream equation for all 1,124 fitted cells (unit tests).

The coefficients β_{ik} form the cell's spike-history filter. Its steady-state gain $\bar{\beta}_i = \frac{1}{K}\sum_k \beta_{ik}/d_k$ (pA Hz^{-1}) is negative for almost every cell, which is spike-frequency adaptation (Benda and Herz, 2003; Weber and Pillow, 2017): median −10.5 for pyramidal (Pyr) cells, −6.6 for SST, −4.1 for VIP and −0.8 for PV cells in the development cohort.

Cells and cohorts. Following the upstream release criterion, we kept mouse cells with validation explained variance ≥ 0.5. We did not select on the held-out test sweeps. Operational labels came from Cre line and dendrite type: spiny cells from excitatory lines are Pyr, and Pvalb-, Sst- and Vip-Cre cells are PV, SST and VIP. This gave 597 cells (339 Pyr, 107 PV, 86 SST, 65 VIP) from 410 donors, mostly in primary visual cortex. Whole donors were assigned to a development cohort (298 cells) or a confirmation cohort (299 cells). The assignment was a seeded, type-stratified greedy procedure. No donor contributes to both cohorts. All tuning and piloting used only the development cohort.

4.2 Networks

A network has $N = 100$ cells, 80 Pyr, 10 PV, 5 SST and 5 VIP, sampled without replacement for each network seed. The input current to cell i is

$$I_{i,t} = I_i^0 + S_i\Big(\textstyle\sum_c W_{ic}^{\mathrm{in}} u_{c,t} + \sum_j s_j |W_{ij}| M_{ij}\, r_{j,t-1}/R_0\Big), \tag{3}$$

with $R_0 = 100\,\mathrm{Hz}$ for every cell. The sign s_j is $+1$ for Pyr and -1 for interneurons (Dale's law). The mask M is drawn from a type-level connection-probability matrix inspired by cortical motifs (Pfeffer et al., 2013; Pi et al., 2013). These are modelling assumptions, not measured values (Table 4). Magnitudes $|W_{ij}|$ are log-normal with mean 1 ($\sigma_{\log} = 0.35$), scaled by $0.6/\langle\text{in-degree}\rangle$, and inhibitory sources are multiplied by 1.5. Input weights are $W^{\mathrm{in}}_{ic} \sim \mathcal{N}(0, 1/5)$. The readout $z_t = \sum_{i\in\mathrm{Pyr}} w_i r_{i,t}/R_0 + b$ uses Pyr cells only.

Operating-point calibration. For each cell, I^0_i is chosen on a 33-point grid in $[0, \sigma_i]$ so that its isolated steady-state rate is closest to 5% of its own fitted ceiling γ_i. The rate is measured in the last third of a 3-s constant-current probe. The input scale is $S_i = 0.25\,\sigma_i$. The procedure uses no task information. Because the target is relative to each cell's ceiling, the resulting baseline rates differ by type (PV higher), by construction.

Identity and wiring controls. *Within-type shuffle*: independent permutations, within each type, of the input filters $a_{i\cdot}$, the history filters $\beta_{i\cdot}$ and the activation tuple (σ, γ, c, b). This keeps marginal distributions and breaks the pairing of a cell's own parameters. *Type-homogeneous*: every parameter is replaced by its type mean. *Cross-type permutation*: whole parameter tuples are permuted across types. *Fast decay*: $d_k \leftarrow \max(d_k, 1-e^{-1})$, so $\tau_k \le 20\,\mathrm{ms}$, with a and β rescaled to keep a/d and β/d, and hence the static response. Unless stated otherwise, each manipulated bank is recalibrated with the same label-free procedure ("matched"). The "reference" calibration keeps the intact bank's I^0, S, as in earlier versions of this project; it is shown as a control. *Degree-preserving rewiring* applies $10|E|$ directed double-edge swaps. It preserves every cell's in- and out-degree, its Dale sign and its multiset of outgoing magnitudes.

4.3 Tasks

Inputs have five channels: two sensory streams (A, B), two cue channels and a go signal. Time is binned at 20 ms: fixation 100 ms, stimulus 200 ms, a silent delay of 100, 200 or 300 ms in training, and a 60-ms response window marked by the go channel. Sensory evidence has signed coherence $\pm\{0.25, 0.5, 1\}$ plus Gaussian noise ($\sigma = 0.25$ per bin). Every batch is exactly balanced over cue, relevant sign and irrelevant sign. *T1-early* (context-dependent decision; Mante et al., 2013): a context cue during fixation and stimulus says which stream is relevant, and the target is the sign of the relevant stream. *T1-late*: the same, but the context cue (60 ms) arrives only after the delay, so both streams must be held. *T2-dms* (delayed match-to-sample, Miller et al., 1996; Masse et al., 2019): stream A is the sample and stream B a simultaneous distractor. A ± 1 probe arrives after the delay, and the target is sign(sample) $\times$ probe, so no input before the probe carries information about the choice. Conflict trials are those whose two streams have opposite signs. Test splits are 512 trials each: in-distribution (ID); unseen delays of 400 and 600 ms (OOD); higher sensory noise ($\sigma = 0.5$); a 40-ms zero-mean distractor ($\sigma = 0.75$) in mid-delay; and all three combined.

4.4 Loss and learning rules

The loss is the binary cross-entropy of z_t at every response bin, averaged within a trial and then over the batch. The choice is the sign of the mean response-window logit. Only $|W|$ (masked, non-negative, at most 2), w and b are plastic.

BPTT. The exact gradient by back-propagation through time (Werbos, 1990) through all filters, the spike-history loop and the recurrent connections. It was checked against central finite differences.

Local rule. The local rule is e-prop (Bellec et al., 2020), which is related to RFLO (Murray, 2019), adapted to GFR cells. It approximates $\partial\mathcal{L}/\partial W_{ij}$ by

$$\hat{g}_{ij} = \sum_t L_{i,t}\,\frac{g'_{i,t}}{R_0}\,\bar{e}_{ij,t}, \qquad L_{i,t} = B_i\,\frac{\partial\mathcal{L}}{\partial z_t}, \tag{4}$$

where B is a fixed random feedback vector (Lillicrap et al., 2016) and g' is the activation slope. The K-dimensional eligibility trace $e_{ij,t} = \partial h_{i,t}/\partial W_{ij}$ follows only cell i's own dynamics, with presynaptic activity held fixed:

$$e^{(k)}_{ij,t} = (1-d_k)\,e^{(k)}_{ij,t-1} + a_{ik}S_i s_j\,\frac{r_{j,t-1}}{R_0} + \underbrace{\beta_{ik}\,g'_{i,t-1}\,\bar{e}_{ij,t-1}}_{\text{spike-history term}}. \tag{5}$$

Every quantity is available at the synapse or the postsynaptic cell, plus one broadcast error signal (Gerstner et al., 2018). *No-β* removes only the spike-history term in Eq. 5; the network dynamics are unchanged. *Symmetric feedback* uses $B = w$ (the readout weights), which isolates the error introduced by ignoring paths through other cells. The quantity in Eq. 5 can be computed forward in time, as a synapse would, or by an equivalent reverse sweep over each cell's own dynamics, which is the network gradient with all paths through other cells removed. The two agree to $< 10^{-10}$ relative error (unit tests). We used the reverse sweep because it is $\sim$17$\times$ faster.

Frozen reservoir. Recurrent weights are fixed, and only w, b are trained, with their exact gradient.

Optimizers. *Adam* (Kingma and Ba, 2015) with global gradient-norm clipping at 1.0 is the primary optimizer. *RMS* is per-parameter RMS normalization without momentum and without clipping (Tieleman and Hinton, 2012), so every update depends only on that synapse's own history. It is an optimizer, not a model of homeostatic synaptic scaling (Turrigiano, 2008). After each update, magnitudes are projected onto $[0, 2]$.

Training and model selection. Each run trains for 3000 updates of 32 trials. Validation uses 256 ID trials every 50 updates, and the checkpoint with the lowest validation BCE (including the untrained network) is kept. Learning rates were chosen on the development cohort only. The grid was $\eta_{\text{rec}} \in \{3\cdot10^{-4}, 10^{-3}, 3\cdot10^{-3}, 10^{-2}\} \times \eta_{\text{out}} \in \{0.03, 0.1, 0.3, 1\}$, three seeds per candidate, and the criterion was mean best validation BCE. This was done separately for each learner, task and optimizer (Table 5). A second stage tried $\eta_{\text{out}} \in \{3, 10\}$ at each learner's selected η_{rec}, because with Adam the frozen reservoir and the no-β rule were still improving at the top of the grid. The final choice minimized the same criterion over both stages; the reservoir's optimum ($\eta_{\text{out}} = 3$) is then inside the range tried. An earlier search with $\eta_{\text{out}} \le 0.1$ had put every learner at its upper edge and was discarded. No test split had been evaluated at any of these points. Controls use the rates of their learner. A network seed fixes the sampled cells, graph, initial weights, feedback vector and every trial, so all conditions are paired.

4.5 Analyses

Population codes. For each phase (stimulus, delay, response), we fitted PCA, ridge probes ($\alpha = 1$) for choice, cue and each stream's sign, and a class-mean choice axis on 128 *analysis-fit* trials. All were evaluated on 128 independent *analysis-evaluation* trials, to avoid double dipping (Kriegeskorte et al., 2009).

Gradient alignment. At each saved checkpoint, on 32 independent diagnostic trials, we computed the cosine between the exact recurrent gradient and each local estimate: with or without β, and with random or symmetric feedback. Per-neuron alignment uses the row of the gradient belonging to postsynaptic cell i. Each cell's *adaptation loop gain* is $|\langle g'_i \rangle \, \bar{\beta}_i|$, the steady-state history gain times the mean activation slope on those trials.

4.6 Statistics and protocol

The unit of inference is the network. We avoid pooling cells or trials as if they were independent (Aarts et al., 2014). Comparisons are paired by seed and use exact two-sided sign-flip tests over all 2^{16} sign assignments, with percentile bootstrap CIs over networks. Families of tests and the Holm correction are as preregistered (Section A.1). The $2\times2\times2$ factorial effects (identity, wiring, learner) are main effects and interactions computed within each network. An identity, wiring or heterogeneity effect is called *practically negligible* when its 90% CI lies within ±5 pp. The protocol, the analysis code and the source hash were locked before the confirmation cohort was used (Supplementary Section A.1). Results outside it are labelled exploratory.

Code and data availability

All code, configurations, the locked protocol and every run's raw outputs (per-trial predictions, learning curves, checkpoints) are in the project repository. The fitted neuron models and cell metadata are those released by Yeung et al. (2025) (AllenInstitute/GRNN, commit `835a535`) under their original licence.

References

E. Aarts, M. Verhage, J. V. Veenvliet, C. V. Dolan, and S. van der Sluis. A solution to dependency: using multilevel analysis to accommodate nested data. *Nature Neuroscience*, 17:491–496, 2014. doi: 10.1038/nn.3648.

G. Bellec, D. Salaj, A. Subramoney, R. Legenstein, and W. Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems 31*, 2018.

G. Bellec, F. Scherr, A. Subramoney, E. Hajek, D. Salaj, R. Legenstein, and W. Maass. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11: 3625, 2020. doi: 10.1038/s41467-020-17236-y.

J. Benda and A. V. M. Herz. A universal model for spike-frequency adaptation. *Neural Computation*, 15(11):2523–2564, 2003.

Y. N. Billeh, B. Cai, S. L. Gratiy, et al. Systematic integration of structural and functional data into multi-scale models of mouse primary visual cortex. *Neuron*, 106(3):388–403, 2020. doi: 10.1016/j.neuron.2020.01.040.

G. Chen, F. Scherr, and W. Maass. A data-based large-scale model for primary visual cortex enables brain-like robust and versatile visual processing. *Science Advances*, 8(44):eabq7592, 2022. doi: 10.1126/sciadv.abq7592.

W. Gerstner, M. Lehmann, V. Liakoni, D. Corneil, and J. Brea. Eligibility traces and plasticity on behavioral time scales: experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12:53, 2018. doi: 10.3389/fncir.2018.00053.

N. W. Gouwens, S. A. Sorensen, J. Berg, et al. Classification of electrophysiological and morphological neuron types in the mouse visual cortex. *Nature Neuroscience*, 22:1182–1195, 2019. doi: 10.1038/s41593-019-0417-0.

D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

N. Kriegeskorte, W. K. Simmons, P. S. F. Bellgowan, and C. I. Baker. Circular analysis in systems neuroscience: the dangers of double dipping. *Nature Neuroscience*, 12:535–540, 2009. doi: 10.1038/nn.2303.

T. P. Lillicrap, D. Cownden, D. B. Tweed, and C. J. Akerman. Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7:13276, 2016. doi: 10.1038/ncomms13276.

V. Mante, D. Sussillo, K. V. Shenoy, and W. T. Newsome. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, 503:78–84, 2013. doi: 10.1038/nature12742.

O. Marschall, K. Cho, and C. Savin. A unified framework of online learning algorithms for training recurrent neural networks. *Journal of Machine Learning Research*, 21(135):1–34, 2020.

N. Y. Masse, G. R. Yang, H. F. Song, X.-J. Wang, and D. J. Freedman. Circuit mechanisms for the maintenance and manipulation of information in working memory. *Nature Neuroscience*, 22: 1159–1167, 2019. doi: 10.1038/s41593-019-0414-3.

E. K. Miller, C. A. Erickson, and R. Desimone. Neural mechanisms of visual working memory in prefrontal cortex of the macaque. *Journal of Neuroscience*, 16(16):5154–5167, 1996.

J. M. Murray. Local online learning in recurrent networks with random feedback. *eLife*, 8:e43299, 2019. doi: 10.7554/eLife.43299.

N. Perez-Nieves, V. C. H. Leung, P. L. Dragotti, and D. F. M. Goodman. Neural heterogeneity promotes robust learning. *Nature Communications*, 12:5791, 2021. doi: 10.1038/s41467-021-26022-3.

C. K. Pfeffer, M. Xue, M. He, Z. J. Huang, and M. Scanziani. Inhibition of inhibition in visual cortex: the logic of connections between molecularly distinct interneurons. *Nature Neuroscience*, 16:1068–1076, 2013. doi: 10.1038/nn.3446.

H.-J. Pi, B. Hangya, D. Kvitsiani, J. I. Sanders, Z. J. Huang, and A. Kepecs. Cortical interneurons that specialize in disinhibitory control. *Nature*, 503:521–524, 2013. doi: 10.1038/nature12676.

D. Salaj, A. Subramoney, C. Kraisnikovic, G. Bellec, R. Legenstein, and W. Maass. Spike frequency adaptation supports network computations on temporally dispersed information. *eLife*, 10: e65459, 2021. doi: 10.7554/eLife.65459.

C. Teeter, R. Iyer, V. Menon, et al. Generalized leaky integrate-and-fire models classify multiple neuron types. *Nature Communications*, 9:709, 2018. doi: 10.1038/s41467-017-02717-4.

T. Tieleman and G. Hinton. Lecture 6.5 – RMSProp: divide the gradient by a running average of its recent magnitude. COURSERA: Neural Networks for Machine Learning, 2012.

G. G. Turrigiano. The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3): 422–435, 2008. doi: 10.1016/j.cell.2008.10.008.

A. I. Weber and J. W. Pillow. Capturing the dynamical repertoire of single neurons with generalized linear models. *Neural Computation*, 29(12):3260–3289, 2017.

P. J. Werbos. Backpropagation through time: what it does and how to do it. *Proceedings of the IEEE*, 78(10):1550–1560, 1990.

R. J. Williams and D. Zipser. A learning algorithm for continually running fully recurrent neural networks. *Neural Computation*, 1(2):270–280, 1989.

C. Yeung, K. Ganjam, Z. Zhang, S. Mihalas, and Z. Lu. A dataset of differentiable biologically-derived single neuron models. bioRxiv preprint, 2025. URL `https://www.biorxiv.org/content/10.64898/2025.12.02.691895`.

A Supplementary information

A.1 Locked protocol

The confirmatory protocol (`docs/PREREGISTRATION_v11.md`) fixed the cohorts, network, tasks, training budget, learning-rate selection, conditions, endpoints, test families and equivalence margin before any confirmation-cohort run. Its SHA-256, the source-code hash and the analysis-script hash are recorded in `docs/LOCK_v11.json` (Table 3).

Table 3: Protocol lock record.

locked at utc	`2026-11-08T20:23:06.054737+00:00`
protocol sha256	`34453091eb6167c1fcb563b9edb4ee73...`
source hash	`6f3ff264daccfdeee2dae6656539f3bf...`
analysis sha256	`705004bdb741278852253c48243748dd...`
confirmation configs sha256	`ecadfeb93d31af97f7390359925c76a1...`

A.2 Connection-probability prior

Table 4: Type-level connection probabilities $P(\text{target} \leftarrow \text{source})$ used to draw the recurrent mask. They are modelling assumptions inspired by Pfeffer et al. (2013); Pi et al. (2013), not measurements for the sampled cells.

target \ source	Pyr	PV	SST	VIP
Pyr	0.20	0.50	0.45	0.05
PV	0.45	0.35	0.30	0.05
SST	0.35	0.05	0.05	0.70
VIP	0.30	0.05	0.40	0.05

A.3 Selected learning rates

Table 5: Learning rates selected on the development cohort by validation BCE (recurrent / readout). The frozen reservoir has no recurrent rate.

task	optimizer	BPTT	local	local (no β)	reservoir
early	adam	0.001 / 1	0.003 / 3	0.001 / 3	– / 3
early	rms	0.001 / 10	0.003 / 0.3	0.0003 / 0.3	– / 0.3
late	adam	0.003 / 1	0.001 / 3	0.001 / 1	– / 3
late	rms	0.003 / 3	0.001 / 1	0.003 / 0.3	– / 0.3
dms	adam	0.003 / 1	0.003 / 1	0.001 / 1	– / 3
dms	rms	0.003 / 0.1	0.001 / 1	0.003 / 0.3	– / 0.3

A.4 Supplementary figures

A.5 Relation to earlier versions of this project

Earlier internal versions compared intact and shuffled cells while keeping the intact bank's calibration. They split cells by specimen (65 donors were shared between cohorts), compared learners after a fixed short budget at which none had converged, and used one task. Those results are not used here. All numbers in this paper come from the runs described in Methods.

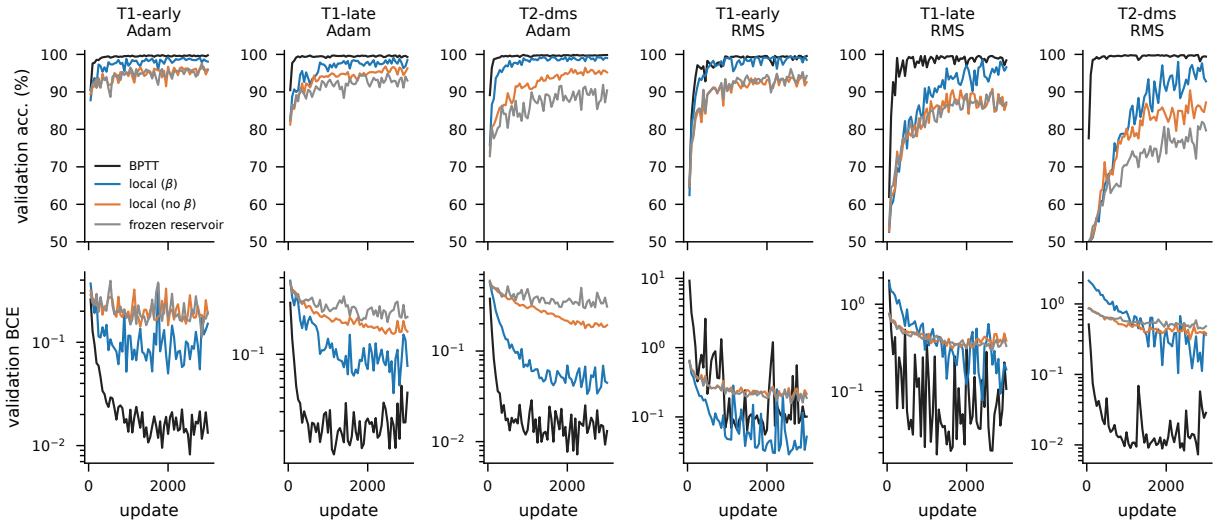


Figure 7: Validation learning curves (in-distribution trials; mean over 16 confirmation networks) for each learner, task and optimizer. Accuracy saturates early, and validation BCE keeps falling for all learners.

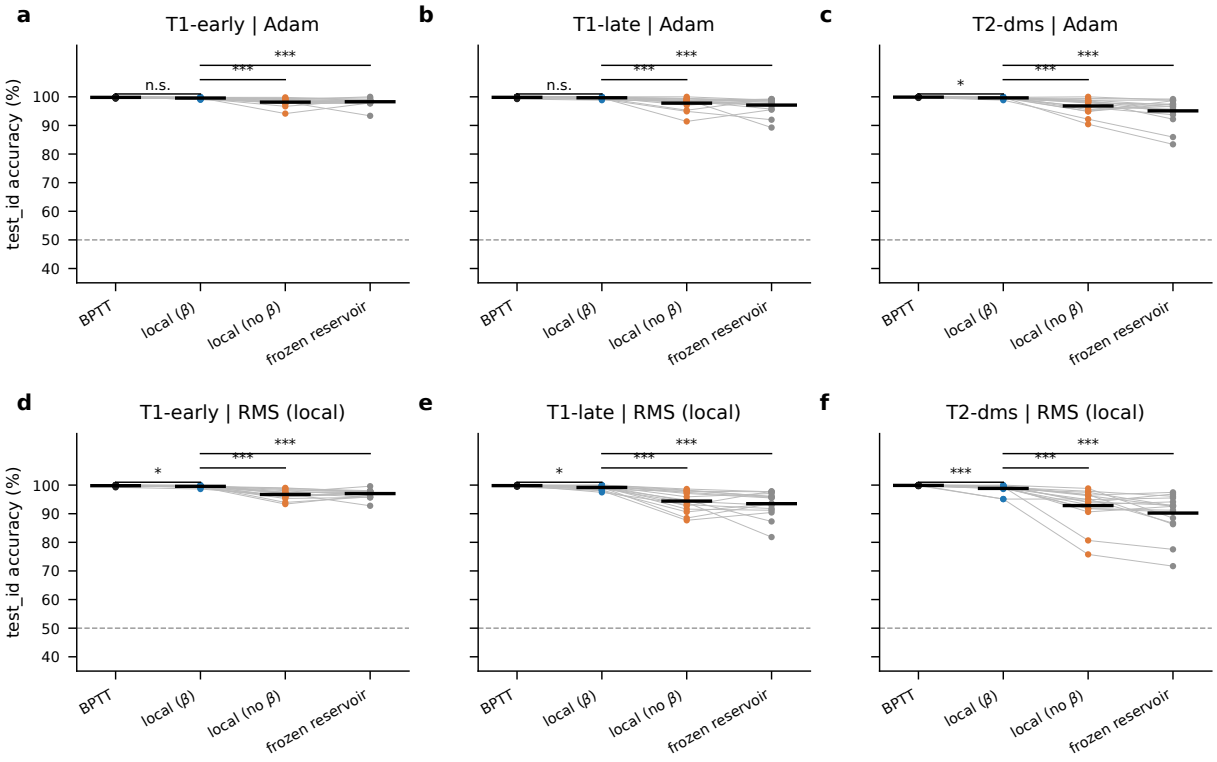


Figure 8: In-distribution accuracy (training delays) for the learners of Fig. 2. Every learner is near ceiling, which is why the preregistered endpoint uses unseen delays. Brackets show the same contrasts on this secondary endpoint (Holm within that endpoint).